

Applications of Formal Languages:

In this essay I will look at 3 examples of applications of formal languages explain what they and how they apply and then finish off with a

Model learning:

Model learning provides a way to understand how to make reliable models. Model learning is the idea that you look at a problem you are trying to solve for a real-world machine for example a coffee maker. Then examine what the inputs and outputs of a system and you record the outputs of the system and systematically put them in a table. Eventually, you can observe that some sort of finite state automaton that explains the behaviour of the system. This method of solving problems can either help identify flaws in a system or help update and improve outdated systems. This way of learning intertwines with machine learning, however, on the surface it can be used for simple real-life machines like a parking ticket machine, a microwave a menu driven self service checkout machine, a vending machine etc. The main relation to formal languages is the inferred grammar or DFA from the data/ table made.

In programming:

This application is more general/ abstract than model learning where code can be more efficient using a finite state machine rather than a less efficient approach. For example, let's say you want to build a traffic light system. The states in the state machine would be red, yellow and green. Then for a two-way system to be implemented if one way has the red state, the other way would have the green state. Then once a certain amount of time has passed the way that is in the green state should transition to orange and then the red state. Then if both states are red, the way that hasn't transitioned, transitions to orange and then green. This then repeats. This way of thinking in can be translated to code and be used to produce and can help solve and improve the performance in many scenarios.

Game development:

One of the major ways state machines are implemented is within game development and even developers who have never heard of the concept have probably come close if not implemented some form of a state machine in their game. This is due to alternate solutions being verbose and taking up too many resources. The reason these are so powerful is the idea of in a game objects can have so many states, and they need to navigate (transition) between states based on certain conditions. For example, the player character in a 2d platformer. The default state of the character would be idle: no buttons pressed by the user inducing a transition in state. Then say the user presses the space bar or w key to jump then the player character would transition to the jump state execute the jump and then if the button is no longer being pressed at the end of the jump execution it would transition back to the idle state. This would be the case for movement state, possible attack state, a dead state being a final state etc. This is only one example of a potential implementation however this could be the case for many facets in any type of game and is why state machines are very powerful tools and a great way of thinking for game development.

Overall, whether it is model learning, general programming, game development or the many other ways state machines, and formal languages can be put to use they are essential tools for any computer scientist that cannot be ignored. Now after the research I have conducted, I will

now be able to identify when a state machine can be a useful tool for me personally as an aspiring software developer in some capacity and in general will include them in my general thinking when solving a problem.

References:

Model learning:

Association for computing machinery (ACM): CAM feb. 2017 – model Learning:

<https://www.youtube.com/watch?v=eg8lKaFmWwM&t=1s>

programming:

This was the one portion I used ai to explore how this was possible and asked it to give me examples of how it can be used. I did this by prompting it to list examples and explain them. I then picked out the traffic light example and explained it in my own words.

Game development:

Bitlytic: finite state machine in Godot 4 in under 10 minutes

https://www.youtube.com/watch?v=ow_Lum-Agbs&t=427s